

Connor M. Mullins

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EDUCATION

The University of Texas at Austin

Expected May 2027

B.S. Cell and Molecular Biology

GPA: 3.85/4.00

Relevant Coursework: Physical Organic Chemistry (Graduate); Molecular Biophysics (Graduate); Advanced Biochemistry (Graduate); Physical Methods in Biochemistry; Molecular Biology; Cell Biology; Microbiology; Engineering Physics; Calculus I–III

RESEARCH EXPERIENCE

McLellan Lab, UT Austin

Sept 2024 – Present

Undergraduate Researcher

- Engineer stabilized HCMV Pentamer and Trimer glycoprotein complexes for structure-based vaccine design using structure-guided and computational approaches (ThermoMPNN, ProteinMPNN); stabilizing substitutions include proline substitutions, disulfide engineering, and cavity-filling mutations; characterize constructs by BLI, SPR, DSF, and negative stain EM.
- Independently collect and process cryo-EM datasets; determined structures of KSHV gH/gL in complex with Fabs to 3.8 Å and HCMV Pentamer with Fabs to 3.6 Å using cryoSPARC.
- Performed AlphaFold-based computational screening of antibody-antigen interactions across MPXV envelope-expressed antigens, identifying OPG153 as a target of broadly neutralizing antibodies; work contributed to publication in *Science Translational Medicine* (2025).
- Developing a novel strategy to determine cryo-EM structures of small proteins (<30 kDa) via covalent nanobody dimerization, exploiting C2 symmetry to overcome preferred orientation and particle size limitations.

Westover Lab, UT Southwestern

May 2024 – Aug 2024

Research Intern

- Crystallized KRAS G12D and KRAS G12D/R68S double-mutant variants; solved X-ray crystal structures to characterize structural mechanisms underlying clinical drug resistance in pancreatic and colorectal cancer.
- Analyzed genome-wide CRISPR dependency data (DepMap) across KRAS alleles in colorectal cancer cell lines; identified allele-specific vulnerabilities including ribosome biogenesis pathway enrichment in G12D lines (GSEA NES = −1.86, FDR < 0.001), consistent with G12D-preferential PI3K/AKT/mTOR signaling. [github.com/conmmul/kras-depmap]
- Expressed and purified a panel of novel KRAS drug-resistance variants generated by lentiviral insertional mutagenesis for biochemical and structural follow-up studies.

PUBLICATIONS

- [1] Townsend, D.R., Zhou, L., Mallory, M.L., Voss, W.N., Knudson, S.A., Acreman, C., Powers, J.M., Hubbard, M.L., Schafer, A., Catanzaro, N., Martinez, D.R., Pareek, S., **Mullins, C.M.**, Nguyen, A.W., Maynard, J.A., Baric, R.S., Georgiou, G., McLellan, J.S., Ippolito, G.C., Lavinder, J.J.
Broad sarbecovirus neutralization by an S2-directed plasma antibody defines a new site of vulnerability in SARS-like viruses. *bioRxiv* 2026.06.06.730562. <https://doi.org/10.64898/2026.06.06.730562>
- [2] Paciello, I., Rundlet, E.J., Zhou, L., Realini, G., Perrone, F., Pierleoni, G., Troisi, M., Postal, J., Guivel-Benhassine, F., Porrot, F., **Mullins, C.M.**, Moschese, D., Cossu, M.V., Sabaini, F., Accordini, S., Ganesini, N., Mantovani, R.P., Panza, F., Fabbiani, M., Tumbarello, M., Antinori, S., Castilletti, C., Montagnani, F., Schwartz, O., Rappuoli, R., McLellan, J.S., Andreano, E..

MANUSCRIPTS

- [1] Zhou, L., **Mullins, C.M.**, McLellan, J.S.
Structure-based engineering of the HCMV Pentamer and Trimer protein complexes. (*in preparation*)
- [2] Zhou, L., Hsieh, C.L., Leist, S.R., Miller, E.H., Horton, A.P., Shoemaker, S.R., Gardner, E.C., Powers, J.M., Boutz, D.R., Mallory, M.L., **Mullins, C.M.**, Johnson, N.V., Tse, A.L., Wang, A., Javanmardi, K., Adams, L.E., Schisler, J., Marqusee, S., Chandran, K., Baric, R.S., Gollihar, J.D., McLellan, J.S.
Structural and functional characterization of a conserved cryptic epitope on SARS-CoV-2 Spike S2 subunit. (*accepted, PLOS Pathogens*)

PRESENTATIONS

- [1] **Mullins, C.M.**, Zhou, L., McLellan, J.S.
Structure-Based Engineering of the HCMV Pentamer and Trimer Protein Complexes. *Capital of Texas Undergraduate Research Conference*, Austin, TX, 2026.
- [2] **Mullins, C.M.**, Zhou, L., McLellan, J.S.
Structure-Based Engineering of the HCMV Pentamer and Trimer Protein Complexes. *UT Undergraduate Research Forum*, Austin, TX, 2026.

HONORS & AWARDS

- **T.W. Whaley Endowed Scholarship** (\$32,000; awarded to ~30 students university-wide) | Cockrell School of Engineering, UT Austin
- **CNS Advanced Summer Research Fellowship** (\$4,000; awarded to 25 students university-wide) | College of Natural Sciences, UT Austin
- **Patricia L. Johnston Biological Sciences Research Fund** | UT Austin
- **Thomas and Elizabeth Merner Scholarship in Natural Sciences** | UT Austin
- **University Honors** | UT Austin
- **Electrical and Computer Engineering Honors** (completed prior to major change) | UT Austin

SERVICE

Undergraduate Research Panelist, Office of Experiential Learning, UT Austin 2025
Invited speaker at the annual university-wide Research Week; addressed 200+ students on navigating undergraduate research opportunities and applying to graduate programs.

Longhorns Crushing Medical Debt 2024–2025
Co-led community workshops educating patients on medical debt relief strategies under the Affordable Care Act; assisted individuals in completing and submitting applications for financial aid.

SKILLS

Computational & Data Analysis: Python (primary), R, MATLAB; pandas, SciPy, NumPy, statsmodels, Matplotlib; RNA-seq analysis; AlphaFold2; cryoSPARC; model building (Coot, ISOLDE); structure refinement (Phenix)

Structural Biology & Biophysics: Cryo-EM data collection and processing; negative stain EM; X-ray crystallography; BLI; SPR; DSF; mass photometry

Molecular Biology: Recombinant protein expression and purification; mammalian cell culture; protein engineering